AMTEC
AGRICULTURAL MACHINERY TESTING AND EVALUATION CENTER

Test Report No. 2023 – 0052

Agricultural Machinery – Furrower
ACT 3-ROW FURROWER

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Test Report Number	2023 – 0052
Date of Issuance	February 23, 2023
Valid Until	February 23, 2028
Agricultural Machinery	Furrower
Type	Tractor-mounted, Three shanks, General-purpose bottom
Brand	ACT
Model	3-ROW FURROWER
Test Requested by	ACT (MACHINERIES AND METAL CRAFT) CORPORATION Turayong, Cauayan City, Isabela

SUMMARY

Performance Criteria	Requirement as per PAES 134:2004	AMTEC Test
Depth of Furrow, mm, minimum	ND[a]	121 ± 13.06
Width of Furrow, mm, minimum	ND[a]	692 ± 32.10

[a] Not specified by the requesting party.
Note: ND – No Data

OBSERVATIONS

1. Operator's manual (based on PAES 102:2000)	
a. Language	English
b. Safety and warnings	Provided
c. Operating information	Provided
d. Accessories and attachments	Not provided
e. Maintenance instruction	Provided
f. Storage	Provided
g. Handling, reception, transportation, assembly, and installation	Provided
h. Specifications	Provided
i. Dismantling and disposal	Provided
j. Warranty	Not provided
k. Parts list	Provided
2. Previous test reports of similar brand and model	None
3. Manufacturer	ACT (MACHINERIES AND METAL CRAFT) CORPORATION
4. Four-wheel tractor used	29.90 kW MASSEY FERGUSON MF1540
5. Number of operators	One
6. Handling and stability when machine is working	The furrower is stable and maintains a smooth operation while it is hitched to the tractor. There were no issues encountered as it maneuver at corners.
7. Adjustments of parts	The depth of furrow can be adjusted using the three-point link of the four-wheel tractor. The furrow spacing can be adjusted with the aid of tools.
8. Straightness of furrows	The straightness of furrows is dependent on the operator.
9. Evenness of spacing between furrows	The spacing between furrows is not measured.
10. Uniformity of depth and width	121 mm ± 13.06 mm (Depth) 692 mm ± 32.10 mm (Width)
11. Durability of parts (based on wear of soil-working parts, visible deformation, etc)	No deformations were observed during the test operation.
12. Safety features	None
13. Failures or abnormalities that may be observed on the machine or its components during and after the operation	No failures or abnormalities observed during the testing operation.
14. The size of the furrower bottom shall be determined by measuring the horizontal distance from the left to the right wings of share.	The size of the furrower bottom is 69 mm.

OBSERVATIONS (Cont'n)

15. The number of furrower bottoms shall depend on crop's row spacing and the tractor's wheel tread adjustment.	The implement has three furrower bottoms with row spacing of 710 mm.
16. The height of the standards shall vary according to the crop requirement but shall not exceed 500 mm.	The height of the standard is 618 mm.
17. The length of the toolbar shall vary according to crop's row spacing and the tractor's wheel tread adjustment.	The length of the toolbar is 1810 mm.
18. The number of toolbars may be one or two depending on the manufacturer's design.	The furrower had a single toolbar.
19. Cast iron and/or mild steel shall be used in the manufacture of the moldboard, standard and toolbar.	The materials of the moldboard, standard, and toolbar were not verified.
20. Carbon steel with at least 80% carbon content (e.g. AISI 1080) or alloy steel with at least 0.0005% boron content shall be used in the manufacture of share.	It was not verified if carbon steel with at least 80% carbon content (e.g. AISI 1080) or alloy steel with at least 0.0005% boron content was used in the manufacture of share.
21. The hitch of the furrower shall be compatible with the three-point linkage of the four-wheel tractor as specified in PAES 118.	The hitch of the furrower is compatible with the three-point linkage of the four-wheel tractor used.
22. The furrower shall be easy to hitch to and unhitch from the tractor as well as adjust the spacing between furrows.	Hitching and unhitching were easy. Adjusting the furrower spacing can be done by loosening the bolts and nuts used to attach the shanks to the toolbars.
23. The furrower shall be free from manufacturing defects such as sharp edges and surfaces that may be detrimental to the operator.	No manufacturing defects observed during the test operation.
24. Except for furrower bottoms, other uncoated metallic surfaces shall be free from rust and shall be painted properly.	All parts are painted, including the furrower bottoms.
25. Other remarks	No breakdown occurred during the test operation.

PURPOSE AND SCOPE OF TEST

Purpose	Performance Testing / Commercial
Date of Test	February 09, 2023
Location of Test	Brgy. Villa Miguel, Quirino, Isabela 17°08'14.83" N, 121°43'8.81" E
Items Tested	Operator's Manual, Depth of Furrow, Width of Furrow, Field Capacity, Field Efficiency, Noise Level, and Fuel Consumption

METHODS OF TEST

The furrower was tested based on the PAES 135:2004 – Agricultural Machinery – Furrower – Methods of Test.

DESCRIPTION OF THE MACHINE

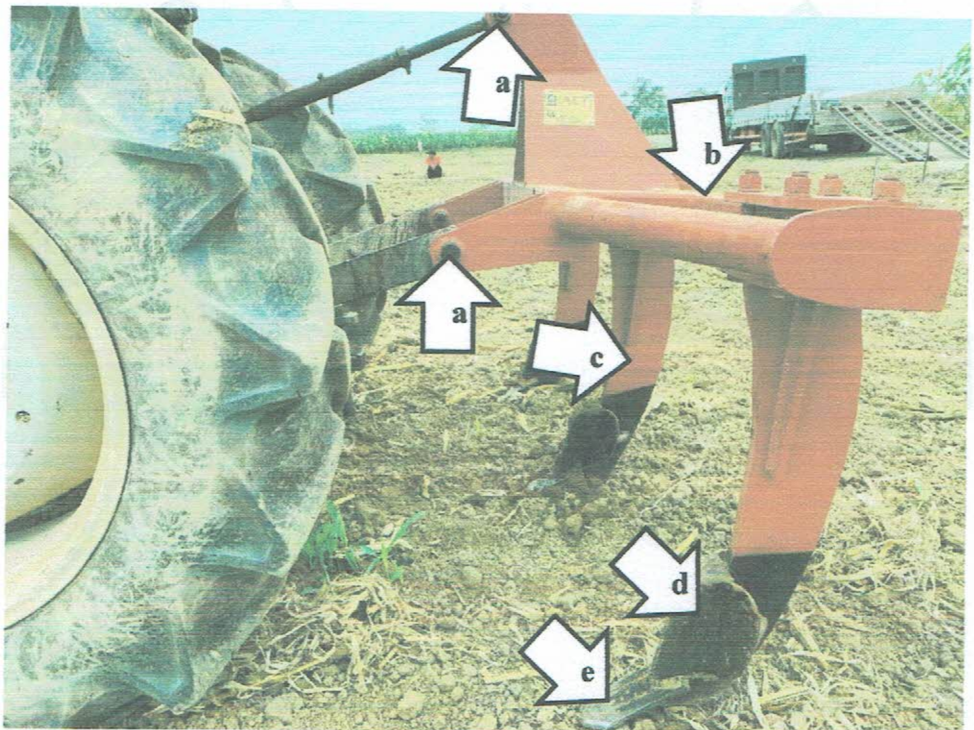

Figure 1. Parts of the ACT 3-ROW FURROWER Furrower include the following: (a) Hitch points, (b) Toolbar, (c) Standard, (d) Moldboard, and (e) Ripper point.

SPECIFICATIONS

Item		Manufacturer's Specification[a]	AMTEC Verification
1	Overall dimensions and weight		
1.1	Length, mm	1803	1810
1.2	Width, mm	530	539
1.3	Height, mm	1060	1068
1.4	Weight, kg	130	130
2	Furrower bottom		
2.1	Quantity	3	3
2.2	Size, mm	140	69
2.3	Material	Allow steel	ND
3	Standard		
3.1	Dimension, L × W × t, mm	605 × ND × ND	618 × 134 × 19.17
3.2	Material	MS plate (t = 19 mm)	Mild steel
4	Toolbar		
4.1	Dimension, L × t, mm	1803 × ND	1810 × 6.20
4.2	Material	Square Tube (76.2 mm × 76.2 mm × 5 mm)	Tubular steel (75 mm × 75 mm)
5	Features		
5.1	Recommended furrow spacing, mm	710	710
5.1.1	Adjustment	Adjustable	Adjustable
5.2	Type of crops	Corn	Corn
6	Operating width, mm	ND	2130
6.1	Adjustment	Adjustable	Adjustable

[a] The data were obtained from the operator's manual and specification sheet provided by the requesting party.
Note: ND – No Data, NM – Not Measured

RESULTS

Item		Data
1	Test Conditions	
1.1	Dimensions of field, m	
1.1.1	Length	45.0
1.1.2	Width	22.5
1.2	Area, m^2	1013
1.3	Soil type	Guingua silty clay loam
1.4	Soil moisture content, %	9.79
1.5	Last crop planted	Corn
1.6	Field condition before operation	Harrowed
2	Field Performance	
2.1	Type of field operation	Secondary
2.2	Tractor's gearshift setting	1st Gear - High
2.3	Traveling speed, kph	5.98
2.4	Average depth of furrow, mm	121
2.5	Average width of furrow, mm	692
2.6	Actual field capacity	
2.6.1	ha/h	0.903
2.6.2	h/ha	1.11
2.7	Theoretical field capacity	
2.7.1	ha/h	1.27
2.7.2	h/ha	0.786
2.8	Field efficiency, %	70.90
2.9	Noise at operator's ear level, dB(A)	86.83
2.10	Wheel slip, %	2.14
2.11	Fuel consumption	
2.11.1	L/h	1.98
2.11.2	L/ha	2.21
2.12	Method of operation	Alternating Headland Pattern

OBSERVATIONS

Figure 2. Quality of work

Figure 3. ACT 3-ROW Furrower nameplate.

Tested by:

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Test Engineer

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MAR 03 2023
Date

REMINDER

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